Before you hit the forward button on your smartphone app next time, pause for a minute. That irresistibly funny image or suggestive video you are sharing may look convincingly real but could easily be a deepfake. Like the one of actress Rashmika Mandanna featured on the previous spread. You see her in a black outfit stepping into an elevator. What it actually is her face swap on a video of British-Indian influencer Zara Patel, posted only a few weeks earlier. Explanation is pointless, the video has already gone viral. "Something like this is honestly extremely scary not only for me but also for each one of us who today is vulnerable to so much harm because of how technology is being misused," the actress said. "...if this happened to me in school or college, I genuinely can't imagine how I could tackle this. We need to address this as a community and with urgency before more of us are affected by such identity theft."

By this time, it wasn't just Rashmika who had fallen victim to the menace. Fake images of actresses Katrina Kaif and Kajol emerged in quick succession—a pointer to the disturbing fact of deepfakes proliferating. "Yes, this is a strong case for legal," is how Big B reacted on November 5 on X (formerly Twitter), after a fact-checker debunked the Rashmika clip as fake. Just days later, however, a video clip from Amitabh Bachchan's own quiz show, *Kaun Banega Crorepati,* pushing poll propaganda in Madhya Pradesh started doing the rounds. Manipulated with a voice dub, it became one of the many fakes circulating this election season.

THE ORIGIN OF DEEPFAKES

Face swapping itself is not anything new. Image editing software like Photoshop has been around since the nineties, enabling anyone with a computer to manipulate images. But the advent of deep learning techniques from 2016 onwards altered the game dramatically. Suddenly, you did not need professionals sitting in well-equipped studios for computer graphics imagery or audio recording; you could do it in real-time working out of a garage. All you needed was a powerful graphics processing unit, of the sort gaming computers run on. And powerful artificial intelligence (AI) available online via various machine learning models. Trained on tonnes of visual data, today's AI models can effortlessly produce new visuals that are virtually indistinguishable from the real ones. Which explains why pornography is the biggest lure for deepfake creators. In fact, that's how the term deepfake originated in late 2017—from a user name that posted pornographic videos on the social media platform Reddit, with the faces of Hollywood celebrities such as Gal Gadot, Scarlett Johansson and Taylor Swift swapped onto existing porn clips.

"Deepfakes are definitely one of the big here-and-now problems of AI," says London-based technology expert, author and start-up mentor Jaspreet Bindra. And as Rajeev Chan-

WHAT ARE DEEPFAKES

Deepfakes, as the name suggests, are deceptive images or audiovisuals created using AI and 'deep learning', where an algorithm is trained on original material and it then learns to produce something that closely resembles that original content. A picture or video of a person, for example, can be seamlessly doctored to look like a different person. Even audio can be manipulated to mimic a person's voice

